

Compact Toeplitz Operators on the Full Weighted L^p Space: A Dual-Weight Characterization

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Claimed status: full solution of Open Question 1.13 in arXiv:2107.03457, subject to human review. For every $1 < p < \infty$ and every Békollè–Bonami weight $\sigma \in B_p$, the criterion below characterizes all bounded symbols u for which $T_u = PM_u$ is compact on the *full* space L_σ^p .

1 Source question

Stockdale and Wagner ask the following on PDF page 8 of [1]:

Open Question 1.13. Given $p \in (1, \infty)$ and $\sigma \in B_p$, characterize the symbols $u \in L^\infty$ for which T_u is compact on L_σ^p .

setting. Theorem 1.8, Theorem 1.9, and Corollary 1.10 extend to simple domains, since in this situation the domain can be equipped with a dyadic structure and the Bergman projection is pointwise bounded by a suitable dyadic majorant, see [21].

We now list some open questions connected to this work.

Open Question 1.13. Given $p \in (1, \infty)$ and $\sigma \in B_p$, characterize the symbols $u \in L^\infty$ for which T_u is compact on L_σ^p , where $1 < p < \infty$ and $\sigma \in B_p$.

Open Question 1.14. Answer Open Question 1.13 with L_σ^p replaced with $\mathcal{A}_\sigma^p$.

Open Question 1.15. Given $p \in (1, \infty)$ and $u \in L^\infty$, characterize the weights σ for which T_u acts boundedly on L_σ^p . Do the same for bounded T_u from L_σ^p to $L_\sigma^{p,\infty}$ for $p \in [1, \infty)$.

Open Question 1.16. Determine the optimal dependence of $\|P\|_{L_\sigma^1 \rightarrow L_\sigma^{1,\infty}}$ on $[\sigma]_{B_1}$.

Corollary 1.4 answers Open Question 1.13 in the case that u is continuous near the boundary. Theorem 1.3 completes the characterization sought in Open Question 1.14 for the subclass of B_p weights satisfying the reverse Hölder inequality. Theorem 1.8 and Theorem 1.9 give sufficient conditions for Open Question 1.15 in the cases $p \in (1, \infty)$ and $p = 1$, respectively. Corollary 1.10 addresses Open Question 1.16 for the subclass of B_1 weights satisfying the reverse Hölder inequality.

The remainder of this paper is organized as follows. In Section 2, we discuss the dyadic

Figure 1: The source question and its surrounding list on PDF page 8 of arXiv:2107.03457.

Let V be normalized Lebesgue measure on the unit ball $\mathbb{B}_n \subset \mathbb{C}^n$. The unweighted Bergman projection is denoted by P , and $T_u = PM_u$. For a weight ρ , write

$$\|f\|_{L_\rho^s}^s = \int_{\mathbb{B}_n} |f|^s \rho dV, \quad \mathcal{A}_\rho^s = L_\rho^s \cap \text{Hol}(\mathbb{B}_n).$$

For a fixed $r > 0$, let $D(a, r)$ denote the Bergman-metric ball of radius r centered at a .

2 Answer

Theorem 1 (Full symbol characterization). *Let $1 < p < \infty$, let $q = p/(p-1)$, and let $\sigma \in B_p$. Put*

$$\tau = \sigma^{1-q} = \sigma^{-1/(p-1)} \in B_q.$$

For $u \in L^\infty(\mathbb{B}_n)$ and any fixed $r > 0$, the following are equivalent.

- (i) $T_u = PM_u$ is compact on L_σ^p .
- (ii) Multiplication by $\bar{u}$ is compact from $\mathcal{A}_\tau^q$ to L_τ^q .
- (iii) The dual-weight local means vanish at the boundary:

$$\lim_{|a| \rightarrow 1^-} \frac{1}{\tau(D(a, r))} \int_{D(a, r)} |u(z)|^q \tau(z) dV(z) = 0. \quad (1)$$

The truth of (1) is independent of the chosen fixed radius $r > 0$.

Idea of the proof

The full L_σ^p domain lets us eliminate all cancellation in the symbol. Pair L_σ^p with its natural dual L_τ^q using unweighted integration. The Hermitian Bergman kernel then gives the exact adjoint formula

$$T_u^* = M_{\bar{u}}P.$$

Because P is a bounded projection onto $\mathcal{A}_\tau^q$, compactness of this adjoint is exactly compactness of multiplication by $\bar{u}$ on the dual analytic space. That multiplication map is the canonical embedding of $\mathcal{A}_\tau^q$ into the measure $|u|^q \tau dV$. A weighted compact Carleson lemma turns compactness of this embedding into the local-density condition (1).

The appearance of the *dual* weight τ , rather than σ , is forced by the Banach adjoint. It is also what makes phase cancellation irrelevant: only $|u|^q$ remains.

3 A compact Carleson lemma for Békollé weights

We give the embedding argument to keep the result independent of any unstated regularity of the weight. We use only standard geometry of fixed Bergman balls and the B_s condition.

Lemma 2 (Weighted compact embedding). *Let $1 < s < \infty$, let $\rho \in B_s$, and let μ be a finite positive Borel measure on $\mathbb{B}_n$ satisfying $\mu \leq C_0 \rho dV$. For any fixed $r > 0$, the inclusion*

$$J_\mu : \mathcal{A}_\rho^s \longrightarrow L^s(\mu)$$

is compact if and only if

$$\frac{\mu(D(a, r))}{\rho(D(a, r))} \longrightarrow 0 \quad (|a| \rightarrow 1^-). \quad (2)$$

Proof. We first record three consequences of $\rho \in B_s$. The tent definition of B_s , together with the fact that a fixed-radius Bergman ball is contained in a Carleson tent of comparable volume, gives the ball estimate

$$\langle \rho \rangle_D \left\langle \rho^{-1/(s-1)} \right\rangle_D^{s-1} \leq C_{s,r,\rho} \quad (3)$$

for every ball $D = D(a, r)$ and for every fixed enlargement of such a ball. Hölder's inequality and (3) imply the subset estimate

$$\frac{\rho(E)}{\rho(D)} \geq c_{s,r,\rho} \left(\frac{V(E)}{V(D)} \right)^s, \quad E \subseteq D. \quad (4)$$

In particular, ρ is doubling along fixed-factor enlargements of Bergman balls and comparable Carleson tents.

The holomorphic submean inequality followed by Hölder and (3) yields, for fixed $0 < r < R$,

$$\sup_{z \in D(a,r)} |f(z)|^s \leq \frac{C}{\rho(D(a,R))} \int_{D(a,R)} |f|^s \rho dV \quad (f \in \mathcal{A}_\rho^s). \quad (5)$$

Assume (2). Choose an r -lattice $\{a_j\}$ for the Bergman metric. The balls $D(a_j, r)$ cover $\mathbb{B}_n$, while fixed enlargements $D(a_j, R)$ have bounded overlap. Applying (5) on the balls meeting $\{|z| > t\}$ gives

$$\int_{|z|>t} |f|^s d\mu \leq C \sup_{|a_j|>t'} \frac{\mu(D(a_j, r))}{\rho(D(a_j, r))} \sum_j \int_{D(a_j, R)} |f|^s \rho dV \quad (6)$$

$$\leq C \sup_{|a|>t'} \frac{\mu(D(a, r))}{\rho(D(a, r))} \|f\|_{\mathcal{A}_\rho^s}^s, \quad (7)$$

where $t' \rightarrow 1$ as $t \rightarrow 1$. Thus the norm of the outer restriction tends to zero. On a fixed compact subball, bounded subsets of $\mathcal{A}_\rho^s$ are normal by (5); Montel's theorem and $\mu \leq C_0 \rho dV$ show that restriction there is compact. Hence J_μ is a norm limit of compact restrictions and is compact.

Conversely, suppose J_μ is compact. Fix an integer $N > n + 1$ and set

$$F_a(z) = \left(\frac{1 - |a|^2}{1 - \langle z, a \rangle} \right)^N, \quad f_a = \frac{F_a}{\rho(D(a, r))^{1/s}}.$$

On $D(a, r)$, the modulus of F_a is bounded above and below by positive constants depending only on N, r . We claim

$$\int_{\mathbb{B}_n} |F_a|^s \rho dV \leq C \rho(D(a, r)). \quad (8)$$

Indeed, decompose $\mathbb{B}_n$ into the nonisotropic annuli on which $|1 - \langle z, a \rangle|$ is comparable to $2^k(1 - |a|^2)$. The k th annulus lies in a Carleson tent whose volume is at most $C2^{k(n+1)}V(D(a, r))$. Applying (4) inside the larger tent shows that its ρ -mass is at most $C2^{ks(n+1)}\rho(D(a, r))$. On that annulus, $|F_a|^s \leq C2^{-kNs}$. Summation gives

$$\int_{\mathbb{B}_n} |F_a|^s \rho dV \leq C \rho(D(a, r)) \sum_{k \geq 0} 2^{-ks(N-n-1)},$$

which proves (8).

Applying the subset argument in the whole unit ball also gives

$$\rho(D(a, r))^{1/s} \geq c V(D(a, r)) \asymp (1 - |a|^2)^{n+1}.$$

Since $N > n + 1$, it follows that $f_a \rightarrow 0$ uniformly on compact subsets as $|a| \rightarrow 1$. The family (f_a) is bounded in the reflexive space $\mathcal{A}_\rho^s$ by (8); bounded point evaluations and the compact convergence show that $f_a \rightarrow 0$ weakly. Compactness of J_μ therefore gives $\|f_a\|_{L^s(\mu)} \rightarrow 0$. The lower bound for $|F_a|$ on $D(a, r)$ yields

$$\frac{\mu(D(a, r))}{\rho(D(a, r))} \leq C \|f_a\|_{L^s(\mu)}^s \rightarrow 0.$$

This proves necessity. Since compactness of J_μ does not refer to a radius, the same equivalence also proves radius independence. $\square$

4 Proof of the characterization

Proof of Theorem 1. The natural Banach dual of L_σ^p , under the unweighted pairing

$$\langle f, g \rangle = \int_{\mathbb{B}_n} f \bar{g} dV,$$

is L_τ^q , where $\tau = \sigma^{1-q}$. The elementary duality of Békollé classes gives $\tau \in B_q$. Hence P is bounded both on L_σ^p and on L_τ^q , and its range in the latter space is $\mathcal{A}_\tau^q$.

First take bounded functions of compact support, so that Fubini's theorem applies directly. Hermitian symmetry of the Bergman kernel gives

$$\langle T_u f, g \rangle = \langle P(uf), g \rangle = \langle uf, Pg \rangle = \langle f, \bar{u} Pg \rangle.$$

Density and boundedness extend this identity to all $f \in L_\sigma^p$ and $g \in L_\tau^q$. Therefore

$$T_u^* = M_{\bar{u}} P \quad \text{on } L_\tau^q. \quad (9)$$

If $M_{\bar{u}} : \mathcal{A}_\tau^q \rightarrow L_\tau^q$ is compact, then (9) factors T_u^* as that compact map after the bounded projection P , so T_u^* is compact. Conversely, if T_u^* is compact, its restriction to $\mathcal{A}_\tau^q$ is compact, and $Pg = g$ on this subspace; the restriction is exactly $M_{\bar{u}}$. Schauder's theorem now gives

$$T_u \text{ compact} \iff M_{\bar{u}} : \mathcal{A}_\tau^q \rightarrow L_\tau^q \text{ compact}.$$

Finally,

$$\|\bar{u}f\|_{L_\tau^q}^q = \int_{\mathbb{B}_n} |f|^q d\mu_u, \quad d\mu_u = |u|^q \tau dV.$$

Since $u \in L^\infty$, $\mu_u \leq \|u\|_\infty^q \tau dV$. Lemma 2, with $s = q$, $\rho = \tau$, and $\mu = \mu_u$, says that this multiplication map is compact exactly when (1) holds. This proves all equivalences. $\square$

5 Consequences and stress tests

Corollary 3 (Unweighted form). *For $\sigma \equiv 1$ and $q = p'$, T_u is compact on $L^p(\mathbb{B}_n)$ if and only if*

$$\lim_{|a| \rightarrow 1^-} \frac{1}{V(D(a, r))} \int_{D(a, r)} |u|^q dV = 0.$$

For $p = 2$, this is equivalently the vanishing of the Berezin transform of the positive symbol $|u|^2$.

Remark 4 (Why pointwise decay is not necessary). *If $u = \mathbf{1}_E$, then T_u is compact precisely when the dual weight of E has vanishing relative density:*

$$\frac{\tau(E \cap D(a, r))}{\tau(D(a, r))} \longrightarrow 0.$$

Thus E may meet every boundary collar, so u need not tend pointwise to zero. This explains why the source's boundary-decay condition is sufficient but does not capture all compact symbols.

Remark 5 (Cancellation audit). *The criterion contains $|u|^q$, not the Berezin transform of u . This is forced by the full L_σ^p domain: inputs can absorb the phase of u . There is no contradiction with the subtler analytic-domain problem, where the input is required to be holomorphic and phase cancellation cannot be removed in this way.*

6 Verification and novelty bounds

The proof was stress-tested in six successive upgrades: the Hilbert-space $p = 2$ factorization, unweighted Banach duality for all p , weighted duality for every B_p weight, the explicit compact-Carleson lemma, sparse-support tests, and an adjacent-literature audit. The key exponent in (8) is strict because $N > n + 1$.

The bounded novelty search covered the run indexes and parsed arXiv corpus, the exact arXiv id and quoted question, the paper’s DOI citation graph in OpenAlex, Crossref searches for compact weighted Bergman embeddings, and formula/keyword searches for compact PM_u operators. Chen [3] explicitly answers the *different* analytic-space Open Question 1.14 for all B_p weights. Tong and Li [2] give general Carleson-measure theory for Békollé-weighted Bergman spaces, consistent with Lemma 2. No source found in this bounded search states the full- L^p_σ dual-weight characterization above. Novelty confidence is therefore moderate, while the validity confidence is high. Human review should focus on the ball/tent passage in (3) and the annular peak estimate (8).

References

- [1] C. B. Stockdale and N. A. Wagner, *Weighted theory of Toeplitz operators on the Bergman space*, Math. Z. **305** (2023), Paper No. 10; arXiv:2107.03457.
- [2] C. Tong and J. Li, *Carleson Measures on the Weighted Bergman Spaces with Békollé Weights*, Chinese Ann. Math. Ser. B **42** (2021), 583–600; doi:10.1007/s11401-021-0280-7.
- [3] J. Chen, *Weighted theory of Toeplitz operators on the Fock spaces*, arXiv:2501.13571 (2025), especially its Bergman-space section answering Open Question 1.14 of [1].